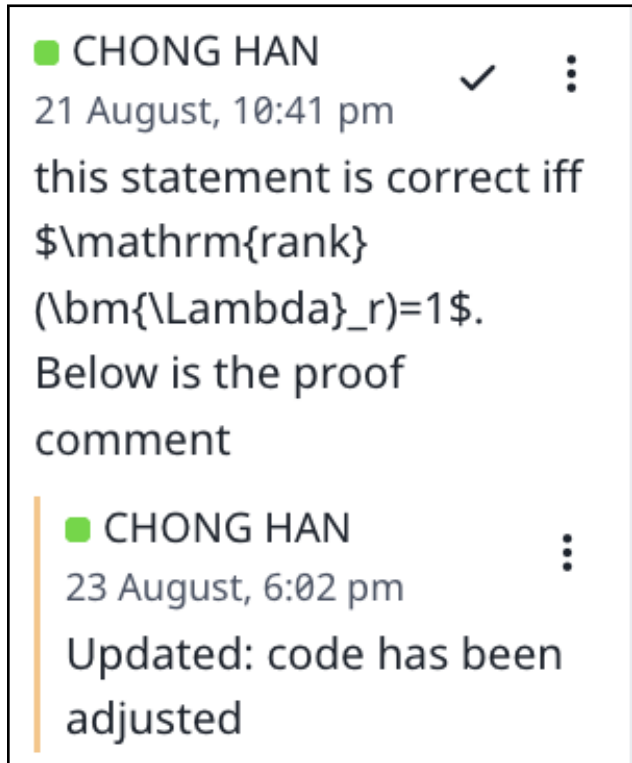


Proposition Proof Review

James Evans

08/23/2025

Proof of Proposition 1 (Cointegration rank $N - 1$)



I agree with the proposed change in this comment. This change is needed to correct the scope of the proof.

Comment 1

I suggest we replace “dimension $N - 1$ when $q_r = 1$ and $\Lambda_r \neq 0$ ” with “dimension $N - 1$ whenever $\mathrm{rank}(\Lambda_r) = 1$.” The rationale is that q_r denotes the dimension of the modeled global nonstationary factor vector g_t in regime r , so it only provides an upper bound ($\mathrm{rank}(\Lambda_r) \leq \min\{N, q_r\}$); the cointegration rank is determined by $\mathrm{rank}(\Lambda_r)$, not by q_r .

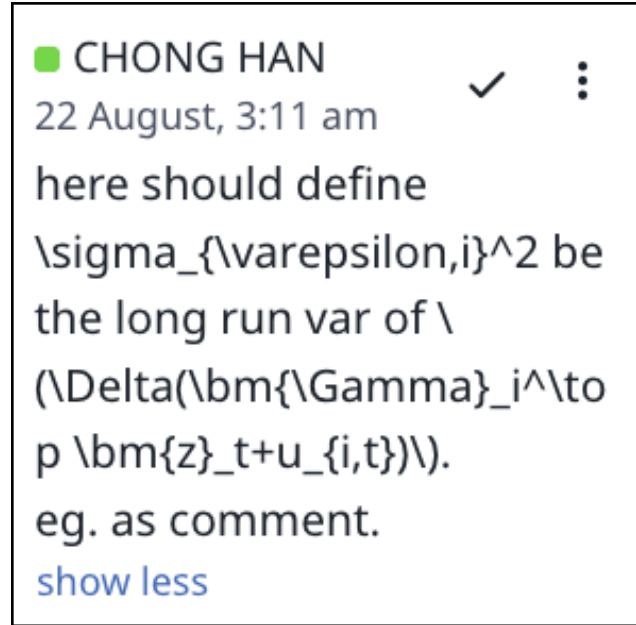
Comment 2

Do we need this proof at all? The general result,

$$\text{cointegration rank} = N - \mathrm{rank}(\Lambda_r),$$

already subsumes it, so this proof appears redundant.

Proof of Proposition 2 (Crisis Co-Movement Amplification of Long Horizon Currency Returns)



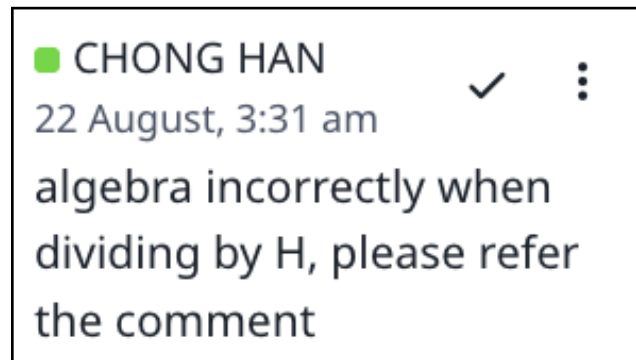
I agree with the proposed change in this comment.

Comment 1

We can implement the long-run variance (LRV) formulation for the stationary remainder as you suggested. However, as a simpler alternative we can instead write

$$\text{Var}(\varepsilon_{i,H}) = O(1) \quad (\text{bounded in } H),$$

because $\varepsilon_{i,H} = \Gamma_i^{\top} (z_{t+H} - z_t) + (u_{i,t+H} - u_{i,t})$ is a difference of stationary processes. Let me know what you would rather do.



I agree with the proposed change in this comment.

Comment 1

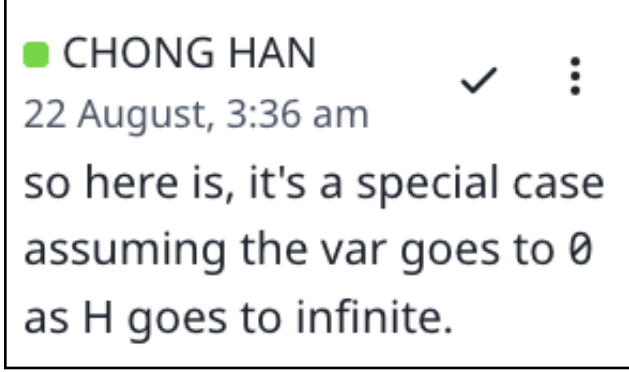
I just wanted to point out that if we went with my definition (bounded remainder), the correction should read:

$$\text{Var}(Y_{i,H}) = H \lambda_{i,r}^{\top} \Sigma_{g,r} \lambda_{i,r} + O(1), \quad (Y_{i,H}, Y_{j,H}) = H \lambda_{i,r}^{\top} \Sigma_{g,r} \lambda_{j,r} + O(1).$$

Therefore,

$$\rho_{ij}^{(H)} = \frac{H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1)}{\sqrt{(H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1))(H \lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1))}} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1/H)}{\sqrt{(\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1/H))(\lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1/H))}}.$$

Let me know which version you prefer to keep.



I agree with the proposed change in this comment.

Comment 1

Under my bounded-remainder definition, $\text{Var}(\varepsilon_{i,H}) = O(1)$ and $(\varepsilon_{i,H}, \varepsilon_{j,H}) = O(1)$. After dividing by H , these contribute only $O(1/H)$ terms, which vanish as $H \rightarrow \infty$. Thus this is not an issue—the idiosyncratic part properly disappears in the limit.